ALEXANDRA ZACH

Portland, OR | 847-212-3858 | allyzach28@gmail.com | allyzach.com | @0xallyzach

WORK EXPERIENCE

Pantera Capital Remote

Research Engineer *April 2024 - October 2026*

- Analyzed $7.8 billion in prediction-market activity and built portfolio-company dashboards using price-impact regression, behavioral segmentation, and cohort analysis.
- Co-developed the Tokenization Progress Index to measure asset maturity; built a public analytics portal reconciling blockchain and issuer data across 600+ assets and co-authored two reports.
- Conducted a 502-response compensation study using statistical power analysis, hypothesis testing, and regression controls to distinguish pay differences from experience effects.
- Automated SEC filing extraction for approximately 30 digital asset treasury companies, building a public dashboard that distinguished asset appreciation from per-share treasury growth and dilution.
- Used regression to assess transaction-fee sensitivity and survey-based behavioral models to evaluate reward size, vesting, and payout timing.
- Benchmarked AI agents against human influencers using engagement and sentiment analysis and evaluated the economics of decentralized AI infrastructure.
- Built an Arbitrum testnet exchange and arbitrage competition to study Timeboost's effects on trading behavior and market mechanics.
- Automated investment-team pitch intake and deal tracking through messaging and database integrations and developed daily AI research briefs.

Messari Remote

Enterprise Research Analyst *April 2022 - April 2024*

- Benchmarked acquisition, cohort retention, and fee revenue across six blockchain networks to distinguish lasting adoption from incentive-driven activity.
- Built SQL and Python pipelines and reusable dashboards for cohort retention, user segmentation, and cross-chain activity analysis.
- Authored more than 50 research reports evaluating startup business models, product adoption, and competitive positioning across crypto infrastructure and applications.
- Mapped cross-application usage and user migration using PageRank and community detection, including a network study of 65 Solana applications.
- Analyzed social-platform engagement, creator monetization, and incentives through user segmentation, cohort analysis, and quantitative modeling.
- Developed graph-based heuristics to identify suspected incentive farming and quantify its effects on retention and revenue measurement.

Magnusson Klemencic Associates Seattle, WA

Structural Engineer *July 2018 - March 2022*

- Designed and analyzed more than 7 million square feet of high-rise buildings for seismic and wind loads across the U.S. and Southeast Asia.
- Built nonlinear 3D models for performance-based design, analyzing simulation results to evaluate building behavior and size structural members.
- Developed firm-wide Python and Excel VBA tools to postprocess model outputs and automate foundation and moment-frame member design.
- Coordinated structural design with architects, contractors, and engineers from concept through construction.

PROJECTS

Syndicate

Technical Cofounder & Sole Engineer

- Built and launched a local-first desktop workspace for persistent AI agent teams on macOS and Windows using Electron and React, integrating Claude Code, OpenAI Codex, Gemini CLI, and GitHub Copilot.
- Engineered an event-driven runtime that turns requests into tracked tasks, dispatches specialist agents, and routes their replies to a manager agent for follow-up, with configurable human approvals.
- Built an AI-guided team builder and agent marketplace with reusable skills, per-agent model and MCP configuration, and a live dashboard for tracking work across agents.
- Automated signed releases and updates and added atomic writes, backup restoration, and crash recovery.

EDUCATION

University of Illinois at Urbana-Champaign

M.S., Civil and Environmental Engineering

B.S., Civil and Environmental Engineering; Minor in Business

Areas of Emphasis: Structural Engineering & Construction Management

Recognitions: James Scholar Honors Program, College of Engineering Dean's List

Awards: Engineering Visionary Scholar, C.S. and Ruth Monnier Scholarship, Corsetti Scholarship Fund